

INVENTION/DISCOVERY	DATE	ORIGINATOR	PLACE OF ORIGIN
Programmable computer	1822	Charles Babbage	Britain
Electromagnet	1823	William Sturgeon	Britain
Internal combustion engine	1826	Samuel Morey	US
Water turbine	1827	Claude Burdin, Benoît Fourneyron	France
Steam locomotive	1829	George Stephenson	Britain
Electrical generator	1831	Michael Faraday	Britain
Refrigerator	1834	Jacob Perkins	US/Britain
Vulcanization of rubber	1837	Charles Goodyear	US
Polystyrene	1839	Eduard Simon	Germany
Undersea telegraph cable	1858	Charles Wheatstone	Britain
Theory of evolution	1859	Charles Darwin	Britain
Pasteurization	1862	Louis Pasteur, Claude Bernard	France
Laws of heredity	1866	Gregor Mendel	Austria
Dynamite	1867	Alfred Nobel	Sweden
Periodic table	1869	Dmitri Mendeleev	Russia
Telephone	1876	Alexander Graham Bell	Britain
Phonograph	1877	Thomas Edison	US
Incandescent light bulb	1878	Joseph Wilson Swan	Britain
Automobile	1885	Karl Benz	Germany
Petrol engine	1886	Gottlieb Daimler	Germany
Wireless communication	1893	Nikolai Tesla	Austria-Hungary
Radio telegraph	1895	Guglielmo Marconi	Italy
Cinematography	1895	Auguste & Louis Lumière	France
Radium	1898	Marie & Pierre Curie	Poland/France
Quantum theory	1900	Max Planck	Germany
Rigid dirigible airship	1900	Ferdinand Graf von Zeppelin	Germany
Airplane (controlled powered flight)	1903	Wright brothers	US
Conditioned reflexes	1904	Ivan Pavlov	Russia
Theory of relativity	1905	Albert Einstein	Switzerland
Bakelite plastic	1909	Leo Baekeland	Belgium
Stainless steel	1913	Harry Brearley	Britain
Structure of the atom	1913	Niels Bohr	Denmark
Television	1925	John Logie Baird	Britain
Law of the expanding universe	1929	Edwin Hubble	US
Nylon	1935	Wallace Carothers	US
RADAR	1935	Robert Watson-Watt	Britain
Jet engine	1937	Frank Whittle	Britain
Ball-point pen	1938	László Bíró	Hungary
Nuclear reactor	1942	Enrico Fermi	Italy /US
Aqualung	1943	Jacques Cousteau, Emile Gagnan	France
Atomic bomb	1945	J. Robert Oppenheimer	US
Photosynthesis	1946	Melvin Calvin	US
Commercial jet airliner	1948	Vickers	Britain
Radiocarbon dating	1949	Willard Libby	US
Big Bang theory	1949	George Gamow, Ralph Alpher, Robert Herman	US
Structure of DNA	1953	Francis Crick, Rosalind Franklin, James D. Watson	Britain/US

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Communications satellite	1958	Kenneth Masterman-Smith	US
LASER	1960	Theodore H. Maiman	US
Plate tectonics	1967	Dan McKenzie	Britain
Microprocessor	1969	Intel	US
Email	1971	Ray Tomlinson	US
Genetic modification	1973	Stanley Norman Cohen, Herbert Boyer	US
Personal computer	1973	Xerox PARC	US
Cellphone	1973	Martin Cooper (Motorola)	US
Compact disc	1980	Philips Electronic/Sony Corporation	Netherlands/Japan
World Wide Web	1990	Tim Berners-Lee	Britain
Global Positioning System	1995	US Department of Defense	US
Genetic cloning	1996	Ian Wilmut, Keith Campbell	Britain
Portable media player	2001	Apple	US

PHILOSOPHY AND RELIGION

The earliest inquiries into the nature and meaning of life come from the founders of the great Eastern religions. Since their time, Western philosophers have journeyed to the outer limits of thought and understanding, posing questions that challenge our most fundamental beliefs.

Originating from almost every corner of the globe, the world’s great faiths are as diverse as its cultures. Some have their origin in prehistoric times, yet the 20th century saw the emergence of several new religions that have attracted followers in their millions.

GREAT THINKERS			
NAME	LIVED	ORIGIN	IDEAS/KEY WORK
Siddhartha Gautama (Buddha)	c. 563–483 BCE	India	Founder of Buddhism as a path to achieving nirvana (spiritual enlightenment) and thus release from the earthly cycle of reincarnation.
Lao Tzu	Active 6th century BCE	China	Founder of Daoism, concerning an individual’s approach to life. <i>Dao De Jing</i> .
Confucius	551–479 BCE	China	Founder of Confucianism: social harmony is promoted via social conventions and practices.
Pythagoras	c. 550–c. 500 BCE	Greece	Polymath interested in esoteric knowledge (that he made available to only a few initiates) and the mystical power of numbers.
Socrates	c. 469–399 BCE	Greece	One of the founders of Western philosophy, to whom this quote is attributed: “A life unexamined is not worth living.” No surviving writings.
Plato	c. 427–347 BCE	Greece	A pupil of Socrates; argued that everything we perceive is a mere shadow of its abstract, ideal Form. <i>The Republic</i> (c. 360 BCE).
Aristotle	384–322 BCE	Greece	Wide-ranging philosopher with a special interest in logical classification. <i>Metaphysics</i> (350 BCE).
Plotinus	205–270	Greece / Roman Empire	Founder of Neoplatonism, a development of Plato’s original ideas. <i>Enneads</i> (c. 253–70).
St. Augustine of Hippo	354–430	North Africa/ Roman Empire	Transmitted Platonism through Christian theology. <i>The City of God</i> (413–26).

